

# Curvilinear Coordinates

## 1 Unit Vectors and Scale Factors

A point in three-dimensional space can be represented by its Cartesian coordinates,  $x_i$ ,  $i = 1, 2, 3$ . For certain applications, however, it can be more convenient to work with other *curvilinear coordinates*,  $u_i$ ,  $i = 1, 2, 3$ . Suppose the Cartesian and the curvilinear coordinates are linked by the transformation

$$x_i = x_i(u_1, u_2, u_3),$$

then for a point  $\mathbf{r} = (x_1, x_2, x_3)$ , we have

$$d\mathbf{r} = \sum_{k=1}^3 \frac{\partial \mathbf{r}}{\partial u_k} du_k.$$

The scale factors and the unit vectors are then defined as

$$h_k = \left| \frac{\partial \mathbf{r}}{\partial u_k} \right| = \sqrt{\sum_{i=1}^3 \left( \frac{\partial x_i}{\partial u_k} \right)^2}, \quad \hat{e}_k = \frac{1}{h_k} \frac{\partial \mathbf{r}}{\partial u_k}.$$

The unit vector  $\hat{e}_k$  is in the direction of increasing  $u_k$ , and is generally not a constant vector. If  $\hat{e}_k \cdot \hat{e}_l = \delta_{kl}$ , then  $u_k$ 's are orthogonal curvilinear coordinates.

## 2 Line, Area, and Volume Elements

The infinitesimal arc length is  $(dl)^2 = d\mathbf{r} \cdot d\mathbf{r}$ . In terms of the orthogonal curvilinear coordinates,

$$(dl)^2 = \sum_{k=1}^3 h_k^2 du_k^2.$$

For vector surface area, consider the 1-component,

$$(d\mathbf{a})_1 \hat{e}_1 = (d\mathbf{r})_2 \hat{e}_2 \times (d\mathbf{r})_3 \hat{e}_3 = (h_2 h_3 du_2 du_3) \hat{e}_1.$$

Similar results can be obtained for the other two components.

For the volume element,

$$dV = (d\mathbf{r})_1 \hat{e}_1 \cdot ((d\mathbf{r})_2 \hat{e}_2 \times (d\mathbf{r})_3 \hat{e}_3) = h_1 h_2 h_3 du_1 du_2 du_3.$$

### 3 Differential Operators

The gradient operator is

$$\nabla f = \sum_{k=1}^3 \frac{1}{h_k} \frac{\partial f}{\partial u_k} \hat{e}_k.$$

The divergence operator is

$$\nabla \cdot \mathbf{A} = \frac{1}{h_1 h_2 h_3} \left\{ \frac{\partial}{\partial u_1} (A_1 h_2 h_3) + \frac{\partial}{\partial u_2} (A_2 h_3 h_1) + \frac{\partial}{\partial u_3} (A_3 h_1 h_2) \right\}.$$

The curl operator can be most conveniently expressed as

$$\nabla \times \mathbf{A} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_1 & h_2 \hat{e}_2 & h_3 \hat{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 A_1 & h_2 A_2 & h_3 A_3 \end{vmatrix}.$$

Finally, the Laplacian is

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[ \frac{\partial}{\partial u_1} \left( \frac{h_2 h_3}{h_1} \frac{\partial f}{\partial u_1} \right) + \frac{\partial}{\partial u_2} \left( \frac{h_3 h_1}{h_2} \frac{\partial f}{\partial u_2} \right) + \frac{\partial}{\partial u_3} \left( \frac{h_1 h_2}{h_3} \frac{\partial f}{\partial u_3} \right) \right].$$